

Wilson, Chow Sin Long

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Senior Product Frontend Engineer · AWS Solutions Architect Associate

SUMMARY

Senior product frontend engineer with 5 years of experience owning React and React Native product flows — from Figma and design systems through frontend development, tRPC APIs, and release. Pays close attention to UI and UX details, not only whether the flow ships. Shipped **15+** production apps.

EXPERIENCE

Frontend Developer, Vioo Company Limited

May 2022 – Present

Primary frontend engineer for concurrent client products, scoping UX with designers and stakeholders, turning Figma into tokenized, composable UI, and owning each product flow through release.

Healthcare AI Agent

2025 – 2026

- Designed the chat and admin UX in **Figma** and owned the client-facing flows — document upload, in-chat questionnaire, product and cart actions.
- Hardened Next.js admin and client apps with **Better Auth**, HttpOnly cookies, CORS, and RBAC.
- Streamed **LangGraph** agent replies into the chat with the **AI SDK** over tRPC, rendering tool calls as product cards, cart actions, and questionnaires instead of plain text.
- Built an admin form builder with conditional logic and ran those questionnaires in-thread, so product recommendations and cart actions never left the conversation.
- Optimized expensive components with **memoization** and **useTransition** for heavy UI updates.
- Used **TanStack Query** with server prefetch and client-side rehydration for data lists, and **TanStack Table** with pagination, filters, and sorting; stored table state in URL search parameters so lists stay shareable and bookmarkable.
- Modeled multi-step health questionnaires with **XState** and **React Hook Form / Zod**.

High-Performance Editorial Platform

2025

Led frontend development for a content and search editorial platform built with Strapi CMS, Meilisearch, and **Next.js**.

- Latest 100 articles statically rendered at build time, leaving the rest incrementally rendered on demand. Strapi **GraphQL** queries tagged properly with **Next.js** cache tags; revalidated through webhook triggered by content update events.
- Shipped instant search feature using Meilisearch with full-text search and fuzzy matching; result hits in **< 50ms**.
- PageSpeed Insights average scores **>90**; Reserved image space to prevent layout shifts and used **caching** and **image optimization** (enforced webp / avif formats and compression to **< 100 KB**) to reduce LCP: **FCP < 0.3s**, **LCP < 0.5s**, **CLS < 0.001**.
- Treated keyboard-accessible navigation, accessible form-error handling, semantic structure, and Lighthouse / axe checks as release requirements rather than post-launch fixes.
- Built a Strapi plugin for AI content generation with AI SDK.

React Native Skia App

2024

- Owned the photobooth app from design and development through release on the Apple App Store and Google Play.
- Users take a selfie, load it onto the canvas, edit with stickers and pen tools, then upload and share.
- Built the canvas with React Native Skia, React Native Reanimated, and gesture handlers; captured images with react-native-vision-camera.
- Converted SVG stickers into Skia SVG React components, enabling resize, rotate, pan, and scale on the canvas.
- Managed state with **Zustand**; kept selections atomic and debounced high-frequency updates.
- Optimized with memoization and context isolation; held **55+ FPS** in RN Perf Monitor with over 100 stickers and strokes on the canvas.

React Native Student App

2023

- Owned student flows (course catalog, booking, announcements) from Figma through release on the Apple App Store and Google Play.
- Maintained **99%** crash-free sessions after launch by diagnosing crashes, device compatibility issues, and API bottlenecks with **Sentry**.
- Data fetched with **Tanstack Query**, cut duplicate fetches with cache and staleTime; client state managed with **Redux Toolkit**.
- Reduced feed scroll jank after profiling identified JS-thread stalls from long lists and full-resolution images; migrated lists to **FlashList** with row memoization, cached and downscaled feed images, and ran **Reanimated** transitions and gesture handlers on the native thread.
- Froze inactive tabs and blurred screens, and deferred mounts, to prevent unnecessary re-renders.
- Routed promo and milestone notifications to the matching screen via Firebase Cloud Messaging deep links.

Across client products:

- Built shared **design systems** on **Radix / Tailwind / shadcn/ui**: themed with CSS **tokens** (color, type, space) instead of per-app hex, and exposed typed, reusable component APIs via React composition — compound components, Slot / children, and shared hooks — rather than prop-heavy leaf components. **Storybook**, Fumadocs, and MDX documented the contract; adoption across client apps eliminated duplicate UI.
- Established **Vitest** and **Playwright / Cypress** as quality gates for shared UI and the product flows using it.
- Instrumented client web and mobile apps and their APIs with **Sentry**, making production crashes, device issues, and API errors visible after release.
- Led adoption of **tRPC** and Scalar-generated API documentation, cutting API integration time by **50%** and eliminating schema-mismatch errors that blocked product work.
- Replaced WordPress across client sites with **headless CMS** platforms (Strapi / Sanity) and **unified search** (Meilisearch / Algolia) on **Next.js**; cut load times by **80%** and search latency by **90%**, and achieved PageSpeed Insights scores above **95**.
- Mentored summer interns and junior developers through pair programming, code reviews, and guidance as they shipped production UI.

OTHER WORK

Also shipped restaurant loyalty/student/membership apps (React Native), a homebound elderly care app (Flutter), campaign / e-commerce storefronts (Next.js, Shopify **GraphQL**), and CMS content sites.

SKILLS

Frontend: TypeScript, React, Next.js, React Native, Tailwind CSS, shadcn/ui, Figma, Zustand, Redux

UI behavior: TanStack Query, TanStack Form, React Hook Form, XState, TanStack Table, next-intl

Architecture: state modeling (URL / server / local), design systems, tokens, React composition, component APIs, routing, SSR/ISR, GraphQL, tRPC, OpenAPI, Storybook, Fumadocs

Backend: Node.js, Hono, Drizzle, Prisma, OpenAPI, tRPC, Postgres, DynamoDB, Better Auth

AWS & DevOps: SST, Docker, GitHub Actions, AWS, Vercel, Cloudflare

Quality: Vitest, Playwright, Cypress, React Testing Library, Lighthouse, axe, Sentry

Mobile: Expo, FlashList, Reanimated, Moti, Skia, Unistyles, FCM

QUALIFICATIONS

- AWS Certified Solutions Architect – Associate, 2026
- AWS Certified Cloud Practitioner, 2025

EDUCATION

University College London

2016 – 2020

- Master of Pharmacy (MPharm, undergraduate) — First Class Honours
- Transitioned from pharmacy to software development after finding passion in building applications.